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Technical Section

# Aniso-GS: Anisotropic appearance field for complex highlight modeling in 3D Gaussian splatting

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ABSTRACT

3D Gaussian splatting (3DGS) has significantly advanced the development of novel-view synthesis in both academic research and industrial applications due to its real-time rendering capability and excellent rendering quality. However, the visual quality of 3DGS is constrained when modeling complex scenes with specular highlights and anisotropic effects due to the limitations of low-order spherical harmonics in capturing high-frequency details. To address this issue, we propose a new method, Aniso-GS, which models the view-dependent appearance of each Gaussian by constructing an anisotropic appearance field. This introduces spatially adaptive spherical harmonic features that map low-order spherical harmonics into higher dimensions, significantly enhancing the model's ability to capture high-frequency details. Additionally, we introduce a multi-view information smoothing strategy that improves generalization to novel viewpoints by smoothing model parameters across different training views. Furthermore, our adaptive densification strategy that suppresses Gaussian growth in low-frequency observation regions while promoting refinement in high-frequency regions, thus improving visual quality without increasing the total number of Gaussians. Quantitative and qualitative results demonstrate that our method achieves state-of-the-art visual quality on five public benchmark datasets. On the challenging Anisotropic Synthetic dataset, the PSNR improves by 5.86 dB compared to vanilla 3DGS, which clearly demonstrates the superior performance of our method in modeling specular highlights and anisotropic effects.

## 1. Introduction

Realistic novel-view synthesis has long been an important research goal in computer graphics and vision, with extensive application potential in robotics, autonomous navigation, medical imaging, cinematography, and virtual/augmented reality. In 2020, the introduction of Neural Radiance Fields (NeRF) [1] by Mildenhall et al. marked a significant milestone in this field. NeRF achieves state-of-the-art (SOTA) novel view synthesis quality by providing a continuous scene representation through a multi-layer perceptron (MLP) and using volume rendering to map 3D scenes to 2D images. However, NeRF's training and rendering speeds have consistently been a bottleneck in its practical application. While subsequent studies have introduced methods such as sparse geometric representations [2,3], low-rank decompositions [4,5], and feature grids [6,7] to accelerate the rendering process, these improvements still fall short of fully addressing the resource-intensive challenges associated with ray marching during volume rendering, resulting in relatively slow rendering times. The introduction of 3D Gaussian Splatting (3DGS) [8] in 2023 has further promoted the rapid advancement of this field, providing high-quality real-time rendering performance and excellent compatibility with GPU-based rasterization pipelines. Specifically, 3DGS models a scene as a set of 3D Gaussians parameterized by position, scale, rotation, opacity, and color, which are then projected to 2D and rendered in real time using a tile-based rasterizer. This method provides an attractive alternative to traditional representations such as voxels, hash grids, and point clouds.

Although 3DGS has several advantages [9], it still faces significant challenges when it comes to modeling complex scenes with specular highlights and anisotropic effects [8,10]. This is primarily due to the limited ability of low-order spherical harmonics (SH) to capture high-frequency scene details. To overcome this limitation, VDGS introduces a NeRF-based appearance field to modulate the color and opacity of Gaussians, thereby improving the continuity of scene appearance representation [11]. Spec-GS uses anisotropic spherical Gaussian (ASG) appearance fields instead of conventional SH to better capture specular

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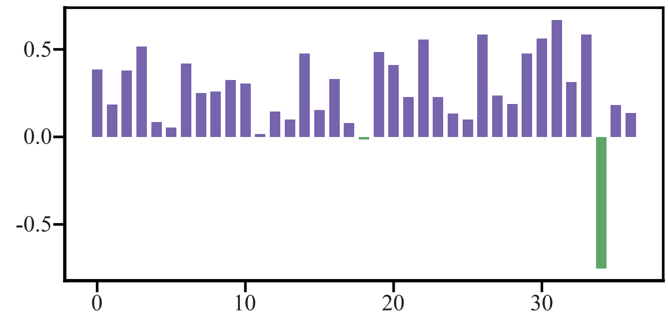


**Fig. 1.** Taking Bonsai in the Mip-NeRF360 dataset [12] as an example, we analyzed the PSNR score differences of the model on each test image before and after multi-view smoothing, and presented the results in bar charts. The results show that the model exhibits local optima on test images with a similar viewpoint to the final training image (as shown in the green bar).

highlights and anisotropic effects [10]. However, these methods still have difficulty accurately reconstructing high-frequency signals.

Furthermore, since 3DGS relies solely on single-view optimization during training, the model is prone to overfitting information from the current view. This increases the likelihood of converging to a local optimum within that view, which consequently undermines the model's overall generalization performance. As shown by the green bar chart in Fig. 1, the model performs well on test images similar to the training view but performs poorly on images with other viewpoints. Additionally, 3DGS uses a uniform gradient statistics method and a fixed densification threshold for all Gaussians by default. This neglects regional variations in Gaussian observation frequency under multi-view observation conditions. In practice, regions with high-frequency observation require higher Gaussian density to enable fine-grained reconstruction of geometric and appearance details. In contrast, regions with low-frequency observation allow for moderate density reduction, thus achieving an effective balance between improving visual quality and controlling memory overhead.

To address these challenges, we propose Aniso-GS, a novel method designed to enhance the ability of the 3DGS to model specular highlights and anisotropic effects, thereby improving visual quality in novel view synthesis. Specifically, we construct an anisotropic appearance field by introducing spatially adaptive spherical harmonic features that map low-order SH into a higher-dimensional representation, significantly strengthening the model's ability to capture high-frequency visual details. We then use an MLP to decode the color and opacity of the Gaussians. Additionally, by smoothing the model parameters across different training views, it effectively avoids the model getting trapped in local optima during training, thereby enhancing its generalization performance in unseen viewpoints. We also introduce an adaptive densification strategy that dynamically adjusts the view-space positional gradients of Gaussians based on their occurrence frequency during training. This strategy suppresses the redundant growth of Gaussians in low-frequency observation regions while promoting refinement in high-frequency observation regions, significantly improving visual quality without increasing the overall number of Gaussians. Comprehensive quantitative and qualitative experimental results demonstrate that our method achieves state-of-the-art performance in novel view synthesis across five public benchmark datasets. On the anisotropic synthesis dataset, our method improves PSNR by 4.86 dB over 3DGS and by 1.98 dB over Spec-GS, respectively, confirming its superior capability in modeling specular highlights and anisotropic effects.

In summary, our main contributions are as follows:

- We propose a novel appearance field based on spatially adaptive spherical harmonic features. By mapping low-order SH into a higher-dimensional representation, this method significantly improves the model's ability to capture high-frequency scene details, thereby improving its ability to represent specular highlights and anisotropic effects.
- We introduce a multi-view information smoothing strategy that effectively prevents the model from converging to local optima during training, thus enhancing its generalization capability to unseen viewpoints.
- Our adaptive densification strategy suppresses Gaussian growth in low-frequency observation regions while promoting refinement in high-frequency regions, thereby improving visual quality in novel view synthesis without significantly increasing the total number of Gaussians.
- We achieve state-of-the-art visual quality on five public benchmark datasets as well as a challenging anisotropic synthetic dataset, demonstrating the effectiveness of our method and its superior performance in modeling specular highlights and anisotropic effects in complex scenes.

## 2. Related work

### 2.1. Neural radiance fields

The emergence of NeRF [1] has significantly advanced the field of novel view synthesis by implicitly modeling the geometry and radiance fields of 3D scenes using MLPs, allowing rendering from arbitrary viewpoints via volume rendering. However, its rendering speed is severely limited by dense ray marching. To address this issue, subsequent research has explored various strategies such as sparse geometry representation [2,3,13–16], low-rank decomposition [4,5,17,18], dense homogeneous voxel grids [6,7,19], and accelerated point search during sampling to improve rendering efficiency. For example, InstantNGP [7] achieves fast feature retrieval using multi-resolution hash coding, which significantly improves rendering speed. In addition, several studies have focused on improving the visual quality of NeRF [20–25] and extending its application to real-world unbounded scenes [12,26,27]. Despite these advances in various aspects, the challenge of resource consumption due to dense ray marching remains unresolved, making high-quality real-time rendering of large scenes particularly challenging.

### 2.2. 3D Gaussian splatting

Based on prior research on modeling object geometry using differentiable points [28–31], 3DGS [8] achieves state-of-the-art visual quality and real-time rendering performance. The proposed method initializes a set of 3D Gaussians from Structure-from-Motion (SFM) [32] points to construct an explicit, high-quality, and unstructured radiance field. It uses differentiable splatting and tile-based rasterization techniques to render images at real-time frame rates. Subsequent studies have optimized 3DGS in terms of anti-aliasing [33], storage compression [34–36], high-fidelity rendering [37], and extended its applications to surface reconstruction [38–42], dynamic scenes [43–45], scene editing [46–48], and 3D generation [49–52].

Although 3DGS achieves high quality results in general scenes, its method of embedding view-dependent effects into individual Gaussians is difficult to adapt to significant viewpoint changes. To address this limitation, Scaffold-GS [34] and VDGS [11] use neural networks to construct Gaussian appearance fields, thereby enhancing the continuous representation of scene appearance. NeRF-GS [53] leverages NeRF's inherent continuous spatial representation capability to overcome the shortcomings of 3DGS in Gaussian initialization, spatial awareness, and inter-Gaussian correlation. This significantly improves reconstruction accuracy and cross-view robustness. However, these methods still struggle to accurately model scenes with specular highlights and anisotropic effects. To tackle this, GShader [54] introduces a lightweight, physics-informed shading function that jointly models the diffuse and specular components of direct illumination, significantly enhancing the appearance fidelity of reflective surfaces. 3iGS [55] decouples emitted radiance into a function of local illumination fields and BRDF features, giving 3DGS structured modeling capabilities for view-dependent reflection behaviors. Finally, Spec-GS [10] observes that traditional low-order spherical harmonics are inadequate for capturing high-frequency signals in scenes, thereby proposing to replace them with anisotropic spherical Gaussians to improve high-frequency signal fidelity in appearance modeling. Based on this, we propose a novel framework that uses spatially adaptive spherical harmonic features to construct a 3D Gaussian anisotropic appearance field, enhancing the modeling capability of 3DGS for complex scenes with specular highlights and anisotropic effects.

Additionally, some methods have attempted to introduce multi-view constraints into 3DGS to improve reconstruction consistency. For example, MVGS [56] departs from the traditional, single-view, independent optimization paradigm by adopting a collaborative, multi-view training strategy. This strategy jointly optimizes Gaussian attributes through cross-view photometric consistency and geometric constraints. Thus, it avoids overfitting to individual training views. Similarly, PGSR [57] improves surface reconstruction accuracy by using pixel correspondence between adjacent views to construct reprojection and normalized cross-correlation losses. However, these existing multi-view approaches typically require synchronous optimization over multiple images, leading to increased computational overhead and training time. In contrast, our proposed multi-view information smoothing strategy only requires single-view input. This avoids getting trapped in local view optima while maintaining the computational efficiency and resource consumption advantages of standard single-view 3DGS training.

Other methods focus on optimizing the adaptive control strategy of 3DGS. GOF [40] and AbsGS [58] identify overly large Gaussians in the blurred regions by improving the calculation method of the view-space position gradient and restore details and alleviate artifacts by performing split operations on the large Gaussians. Pixel-GS [59] proposes to weight the gradient based on the number of pixels covered by each Gaussian from different viewpoints, thereby promoting the growth of large Gaussians. However, a key limitation of these methods is the excessive generation of redundant Gaussians, significantly increasing the GPU memory overhead. In contrast, our proposed adaptive densification strategy suppresses the growth of Gaussians in low-frequency observation regions and refines those in high-frequency observation regions, thereby improving the visual quality of novel view synthesis without significantly increasing the total number of Gaussians.

## 3. Method

The overview of our method is shown in Fig. 2. First, we briefly review the basic concepts of NeRF and 3DGS (Section 3.1). The core of our method is to use spatially adaptive spherical harmonic features to construct a 3D Gaussian anisotropic appearance field for modeling scenes with specular highlights and anisotropic effects (Section 3.2). Next, we introduce a multi-view information smoothing strategy to improve the model's generalization ability (Section 3.3). Finally, we propose an adaptive densification strategy that improves visual quality by suppressing Gaussian growth in low-frequency observation regions and promoting refinement in high-frequency regions (Section 3.4).

### 3.1. Preliminaries

#### 3.1.1. NeRF representation of 3D scenes

NeRF uses the MLP to represent a continuous 3D scene, whose input 5D coordinates consisting of a spatial position $x_n$ and a direction $d_n$, and produces an output of a set of colors $c_n$ and volume densities $\sigma_n$:

$$\mathcal{F}_{\mathrm{NeRF}}(x_n, d_n, \Phi) = (c_n, \sigma_n), \tag{1}$$

where the MLP parameters $\Phi$ are optimized during training by minimizing the photometric difference between rendered and real images. The classical Volume rendering technique is used to accumulate the output colors and volume densities into the 2D image. However, the resource-intensive ray marching process in volume rendering has been a significant bottleneck that limits both the training and rendering speeds of NeRF.

#### 3.1.2. 3D Gaussian splatting

3DGS represents a 3D scene with a set of 3D Gaussians. Each 3D Gaussian is parameterized by a covariance matrix defined in the world coordinate system and the position $u$ (mean) of the SFM [32] point:

$$G(x) = e^{-\frac{1}{2}(x-u)\sum^{-1}(x-u)}, \tag{2}$$

where $x$ denotes a 3D point at any location in the scene. The covariance matrix $\sum = RSS^T R^T$ is parameterized by a diagonal scaling matrix $S$ and a quaternion parameterized rotation matrix $R$ to ensure positive semi-definiteness.

To model the scene appearance, each 3D Gaussian represents the color $c$ using spherical harmonic coefficients and associates an opacity $\sigma$. Then, a tile-based rasterization is designed to efficiently render the image by alpha blending based on the depth order of the primitive:

$$C(x') = \sum_{i\in\mathbf{I}} c_i\varphi_i \prod_{j=1}^{i-1}(1-\varphi_j), \quad \varphi_i = \sigma_i * G'_i(x'), \tag{3}$$

where $G'_i(x')$ denotes the 2D Gaussian obtained from the 3D Gaussian by splatting, and $\mathbf{I}$ denotes the number of ranked 2D Gaussians associated with the query pixel.

Since the entire rendering pipeline is both fast and differentiable, all 3D Gaussian attributes can be directly optimized end-to-end using multi-view photometric loss. During optimization, 3DGS dynamically adjusts the number of 3D Gaussians to more accurately represent the scene. Unlike traditional volume rendering methods that rely on resource-intensive ray marching, 3D Gaussians use tile-based rasterization for efficient scene rendering.

### 3.2. Anisotropic appearance field

Although SH enable view-dependent scene modeling, the inherent low-frequency characteristics of their low-order forms limits their effectiveness in capturing complex optical phenomena such as specular highlights and anisotropic effects [10]. Spec-GS has shown that increasing the SH order only slightly improves performance, but significantly increases the feature dimensions required for each 3D Gaussian, thereby increasing model storage overhead. Therefore, our goal is to construct a latent high-dimensional feature representation for each Gaussian without relying on higher SH orders, enabling more effective encoding of anisotropic information.

Inspired by [10,60], and as shown in Fig. 3, we first use multi-resolution hash grid to encode the Gaussian positions, yielding a feature vector $u \in \mathbb{R}^{32}$. Then, we compute the tensor product of this feature and the direction feature $d \in \mathbb{R}^{16}$ derived from spherical harmonic encoding, and expand it into a high-dimensional feature vector $h \in \mathbb{R}^{512}$, which are referred to as spatially adaptive spherical harmonic features.

$$h = u \odot d, \tag{4}$$

where $\odot$ denotes the tensor product operation, the feature $u$ is a learnable parameter, while the feature $d$ is fixed and unlearnable. In this way, the low-order spherical harmonic coefficients are effectively mapped to a high-dimensional space, significantly enhancing the model's ability to capture high-frequency information. Meanwhile, the feature $u$ can adaptively respond to observation directions with significant luminance changes during training thanks to the structural characteristics of the tensor product, thereby improving the model's

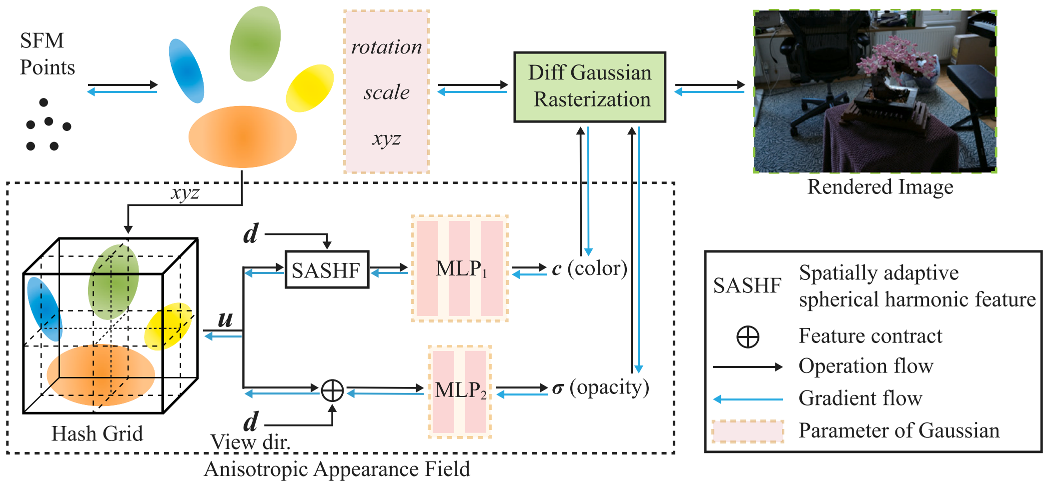


**Fig. 2.** Overview of Aniso-GS. We begin with a sparse point cloud obtained from SFM and generate a set of 3D Gaussians parameterized by position, scale, and rotation. Then, the multi-resolution hash-encoded positional features $u$ and spherical harmonic encoded direction features $d$ are fed into the MLPs to predict opacity and color. Specifically, $\mathrm{MLP}_2$ takes the concatenated position and direction features as input and outputs view-dependent opacity. Meanwhile, SASHF computes the tensor product of the position and direction features, which it then feeds into $\mathrm{MLP}_1$ to generate the color output. Notably, positional features do not participate in the gradient backpropagation process of the anisotropic appearance field.

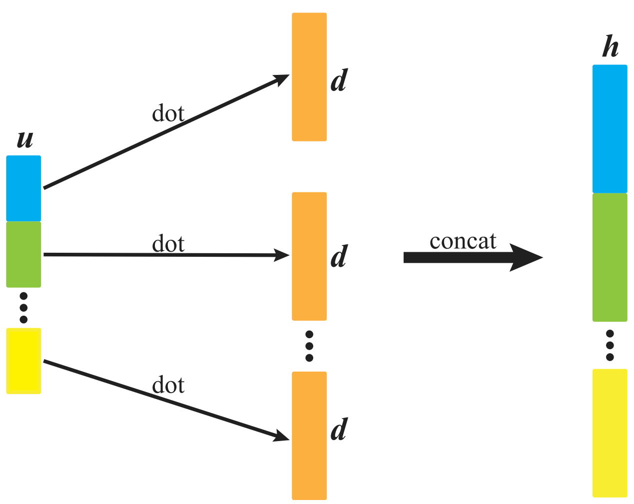


**Fig. 3.** The construction process of spatially adaptive spherical harmonic features. Each component of the position feature $u \in \mathbb{R}^{32}$ is element-wise multiplied with the direction feature $d \in \mathbb{R}^{16}$, and the resulting products are concatenated to form a high-dimensional feature vector $h \in \mathbb{R}^{512}$, thereby enhancing the model's ability to capture high-frequency details.

performance in modeling view-related scenes. Unlike methods such as Spec-GS, this adaptive mechanism does not rely on normal estimation to calculate the reflection direction nor require the construction of complex physical lighting models, making it more concise and universal.

However, when the color response of a Gaussian has low view dependence (i.e., when its color remains approximately isotropic across different views), using spatially adaptive spherical harmonic features to improve its view-dependent modeling capability fails to enhance reconstruction accuracy. In fact, it may lead to overfitting due to excessive model complexity, thereby degrading the visual quality of novel view synthesis. To address this issue, we propose a data-driven view-dependent criterion based on how frequently each Gaussian occurs in multiple views of the training images. Gaussians that occur low-frequently (i.e., in only a few views) tend to have weak view dependency, while those that occur high-frequently are more likely to exhibit significant view dependency. Accordingly, we adaptively enable or disable spatially adaptive spherical harmonic features per Gaussian based on this criterion and decode its color $c$ via a lightweight $\mathrm{MLP}_1$:

$$c = \begin{cases} \mathrm{MLP}_1(h), & f \geq 0.3 \\ \mathrm{MLP}_1(u \oplus d), & \text{otherwise} \end{cases} \tag{5}$$

where $\oplus$ denotes feature concatenation, $f$ denotes the occurrence frequency of a Gaussian across all training views (i.e., the ratio of the number of views in which it is observed to the total number of training views), $\theta$ represents the threshold of the view-dependent criterion and is set to 0.3 by default. When $f < \theta$, the Gaussian is observed in only a minority of views, indicating weak view dependence in its color response. Therefore, we use the default feature concatenation method of NeRF to fuse the position feature $u$ and the direction feature $d$. This frequency-driven adaptive activation strategy significantly improves modeling robustness and generalization in complex real-world scenes. Notably, because the two branches of the $\mathrm{MLP}_1$ network in Eq. (5) receive input features of different dimensions, non-shared $\mathrm{MLP}_1$ instances are used, each configured for its respective input dimensionality.

To further enhance the model's capacity to represent view-dependent appearance, inspired by VDGS, we explicitly model opacity $\sigma$ as a function of the position encoding $u$ and the direction encoding $d$:

$$\sigma = \mathrm{MLP}_2(u \oplus d). \tag{6}$$

This design enables the model to learn and adjust the opacity values from different views, thereby influencing the weight distribution in the alpha blending process and ultimately achieving accurate rendering of view-dependent appearance. Notably, experimental results show that introducing spatially adaptive spherical harmonic features in Eq. (6) does not improve the visual quality of novel view synthesis, so these features are omitted. In summary, we term this comprehensive appearance modeling approach the anisotropic appearance field.

It is worth noting that the optimization of the hash grid comes solely from the gradients of the Gaussian color and opacity transfer. And the gradient of the positional feature $u$ is not backpropagated to the Gaussian position. This prevents the coordinate contraction operation from interfering with the geometric optimization of the 3D Gaussians, thereby avoiding degradation in model convergence performance.

Finally, we parameterize the 3D Gaussians using the opacity and color outputs from the anisotropic appearance field, along with the position, scaling matrix, and rotation matrix initialized from the sparse

SFM point cloud. We then apply Eqs. (2) and (3) for Gaussian splatting and fast rasterization, respectively, to enable real-time image rendering. During model training, we jointly employ the $\mathcal{L}_1$ loss and structural similarity terms $\mathcal{L}_{\text{SSIM}}$ to optimize the learnable parameters and the MLPs.

### 3.3. Multi-view information smoothing

The single-view training strategy in 3DGS causes the model to focus excessively on fitting the current view, which makes it prone to converging to a local optimum. This results in the unbalanced use of multi-view observation data and hinders the accurate reconstruction of the global scene structure. Therefore, during the later stages of training, close to convergence, we introduce the exponential moving average method to smooth the model parameters across consecutive training views, thereby enhancing overall generalization performance:

$$G_t = G_{t-1} * \epsilon + g_t * (1-\epsilon), \quad N - 2 * N_B < t < N \tag{7}$$

where $g_t$ represents the model parameters of the current view, $G_t$ and $G_{(t-1)}$ represent the model parameters after the fusion of the current view and the previous view respectively. $N$ represents the total number of training iterations, and $N_B$ represents the size of the training set. $\epsilon$ is a hyperparameter that regulates the weight distribution between the current and historical view model parameters. It is worth noting that in the early training stage, since the optimization of Gaussian geometry, color, and opacity has not yet converged, the imbalance in multi-view information capture caused by the single-view training strategy does not constitute the primary factor affecting the model's generalization ability. Therefore, we apply multi-view information smoothing only during the final two training cycles, when the model is close to full convergence. The smoothing coefficient $\epsilon$ is set to 0.99 by default in all experiments, and the smoothed model parameters are used during inference.

Multi-view information smoothing helps the model avoid local optima and improves its overall generalization performance under novel viewpoints. As Fig. 1 shows, models with parameter smoothing consistently achieve better test-set performance compared to those without smoothing, which are prone to overfitting the tail view data at the end of training and thus fall into local optima. Additionally, we validate the effectiveness of our method on the vanilla 3DGS.

### 3.4. Adaptive densification strategy

As a critical component of model optimization, the densification strategy in 3DGS enables the transformation of initial 3D Gaussians from a sparse configuration to a denser one, thereby achieving a more accurate representation of the scene's geometric structure. By default, 3DGS computes the average view-space position gradient of visible Gaussians across all training views every 100 iterations. It then applies cloning or splitting operations to the Gaussians whose gradients exceed a predefined threshold. However, due to regional variations in the frequency with which Gaussians occur in multi-view images, a uniform gradient accumulation scheme can cause over-densification in areas with low observation frequency while under-representing regions with high frequency. To address this issue, we propose an adaptive densification strategy that dynamically weights and regulates the view-space position gradients of Gaussians according to their appearance frequencies across views. This mechanism effectively suppresses the growth of Gaussians in low-frequency observation regions while promoting refinement in high-frequency observation regions, thereby improving fine-detail reconstruction in salient regions and enhancing overall visual quality without excessive growth in the total number of Gaussians. Specifically, we initialize the gradient value to zero and use an exponential moving average to accumulate the view-space position gradients of the visible Gaussians in each training view:

$$D_i = \begin{cases} 0, & i = 0 \\ D_{i-1} * \beta + d_i * (1-\beta), & 0 < i < 100 \end{cases} \tag{8}$$

where $i$ denotes the number of iterations, $d_i$ represents the view-space position gradient in the current view, $D_{(i-1)}$ and $D_i$ represent the accumulated view-space position gradient in the previous view and the current view, respectively.

By setting the initial value to zero, Gaussians in low-frequency observation regions exhibit $D_i$ values below the normal level due to fewer iterations. In contrast, Gaussians in high-frequency observation regions undergo more iterations, which allows their $D_i$ values to stabilize and become largely insensitive to initial values. The hyperparameter $\beta$ controls the number of iterations required for $D_i$ to stabilize. A larger $\beta$ makes it more difficult for $D_i$ to overcome the influence of an initial value, necessitating higher observation frequency to restore $D_i$ to its normal level. Therefore, under a uniform gradient threshold, appropriately setting $\beta$ effectively reduces the $D_i$ of Gaussians in low-frequency regions. This suppresses the triggering of cloning or splitting operations and thereby achieving controlled reduction of Gaussian density in these regions. Experimental results show that setting $\beta$ to 0.8 effectively suppress the densification degree in low-frequency area while promoting increased Gaussian density in high-frequency observation areas. This strategy maintains the overall number of Gaussians and improves detail reconstruction quality, achieving an optimal balance between improving visual quality and controlling model storage overhead.

## 4. Experiments

### 4.1. Experimental setup

#### 4.1.1. Datasets and metrics

We evaluated our method on several datasets, including the Synthetic Blender dataset [1], the Shiny Blender dataset [21] (rendered using environment maps), the Mip-NeRF360 dataset [12] with five outdoor real-world and four indoor real-world scenes, as well as Truck and Train from the Tanks&Temples dataset [61] and Drjohnson and Playroom from the DeepBlending dataset [62]. In addition, to evaluate the performance of different methods in handling scenes with specular highlights and anisotropic effects, we conducted experiments on the Anisotropic Synthetic dataset [10] introduced in Spec-GS. Fig. 4 shows that this synthetic dataset comprises eight objects made of typical anisotropic materials, such as brushed metal and CDs, all of which were captured under consistent lighting conditions. To ensure consistent and meaningful performance comparisons, we used common evaluation metrics such as PSNR, SSIM, and LPIPS.

#### 4.1.2. Implementation

We implemented Aniso-GS on the 3DGS repository using the PyTorch framework. The model was trained on a single Tesla V100 GPU with 40 GB of graphics memory. For the neural network architecture, $\text{MLP}_1$ consists of 3 hidden layers, while $\text{MLP}_2$ consists of 2 hidden layers, each with a dimension of 64. The 3D hash grid has a maximum hash table size of 2**19, 16 levels, and a feature dimension of 2. During model optimization, we follow to the same training schedule and hyperparameter settings as in 3DGS, except for the opacity reset operation. The learning rate for grid features is fixed at 0.005, while the learning rates for $\text{MLP}_1$ and $\text{MLP}_2$ are managed using a warmup + exponential decay strategy, with the warmup phase terminated at 10% and 20% of the total training iterations, respectively.


**Fig. 4.** The anisotropic synthetic dataset containing specular highlights and anisotropic effects.

### 4.2. Results and evaluation

#### 4.2.1. Real-world unbounded scenes

We compare with MipNeRF360 [12], 3DGS [8], Mip-Splatting [33], VDGS [11], NeRF-GS [53], Scaffold-GS [34], and Spec-GS [10] methods in real-world scenes to comprehensively evaluate the quality of our approach. The quantitative results are shown in Table 1. It can be seen that our method achieves comparable or even better visual quality than SOTA algorithm on Mip-NeRF360, Tanks&Temples and Deepblending datasets. Compared to Spec-GS, our approach demonstrates greater applicability to real-world scenes. In contrast, the PSNR scores of Spec-GS on the Tanks&Temples and DeepBlending datasets are lower than those of 3DGS, indicating limitations in its reconstruction capability. Furthermore, while our method yields slightly lower reconstruction quality than NeRF-GS on the Tanks&Temples and DeepBlending datasets, it outperforms NeRF-GS across all metrics on the MipNeRF360 dataset. Notably, NeRF-GS requires a two-stage training pipeline. First, a NeRF model is pre-trained to initialize the geometry of the 3DGS. Then, the NeRF and 3DGS representations are jointly optimized. This pipeline significantly increases the training complexity and computational cost [53]. Fig. 5 further demonstrates the visual comparison between our method and prior methods.

As shown in Table 2, we evaluate the rendering speed and storage size of 3DGS, Scaffold-GS, Spec-GS, and our method in terms of efficiency. Although some rendering speed is sacrificed to achieve higher visual quality, our method still supports real-time rendering. Rendering speed is limited by the number of Gaussians in the current view, since the MLP must compute opacity and color for all visible Gaussians. For example, in the Tanks&Temples dataset, the large camera viewpoint results in a large number of Gaussians within the view, which significantly slows down rendering performance. In addition, our storage requirements are drastically reduced compared to 3DGS and comparable to Scaffold-GS due to the use of MLP to output opacity and color. Notably, while Scaffold-GS has advantages in rendering speed, it requires adjusting the voxel size for different datasets. In comparison to Spec-GS, our model achieves lower storage requirements and faster rendering performance.

#### 4.2.2. Synthetic bounded scenes

In addition to real-world scenes, we evaluate our method on the Synthetic Blender dataset and the Anisotropic Synthetic dataset. The Synthetic Blender dataset, known for its exhaustive multi-view images and accurate camera parameters, serves as a widely recognized benchmark. Meanwhile, the Anisotropic Synthetic dataset provides scenes with challenging features such as complex specular and anisotropic highlights. To ensure a fair comparison with prior methods, we use a white background for the Synthetic Blender dataset and a black background for the Anisotropic Synthetic dataset.

As Table 3 show, our method outperforms existing methods on the Synthetic Blender and Anisotropic Synthetic datasets for all three metrics: PSNR, SSIM, and LPIPS. Specifically, our method achieves PSNR improvements of 1.1 and 2.0 dB, respectively, compared to Spec-GS. These results demonstrate that our method more effectively models specular highlights and anisotropic effects, effectively compensating for the shortcomings of 3DGS. For a qualitative comparison of visual quality, see Figs. 6 and 7.

Additionally, we compare our method with 3DGS, GShader [54], and 3iGS [55] on the Shiny Blender dataset, which is rendered using environment maps. As shown in Table 4, our method outperforms the baseline 3DGS and the state-of-the-art 3iGS across all metrics, confirming its enhanced capability to model strongly view-dependent appearance. However, our method achieves marginally lower scores on the Ball and Toaster scenes compared to the results reported in the original GShader paper. This is attributable to the dominance of fine-grained indirect specular reflections on their highly smooth surfaces, which arise from complex environmental lighting. Since our method does not explicitly parameterize or optimize environmental lighting, it has reduced fidelity in reconstructing such indirect reflectance components. GShader, by contrast, incorporates differentiable environment maps and jointly optimizes their representation during training, thereby enabling precise recovery of indirect specular reflection on glossy surfaces. Notably, on the Anisotropic Synthetic and non-reflective Synthetic Blender datasets, our method outperforms GShader across all metrics (see Table 3), indicating stronger generalization capability.

### 4.3. Ablation studies

In this section, we perform ablation studies to evaluate the effectiveness of the following components: spatially adaptive spherical harmonic features (SASHF); view-dependent opacity (V-Opacity); multi-view information smoothing (M-Smoothing); adaptive densification strategy (A-Densify); and frequency-driven adaptive activation strategy (A-Activation). These experiments are completed on both the real-world Mip-NeRF360 dataset and the challenging Anisotropic Synthetic dataset.

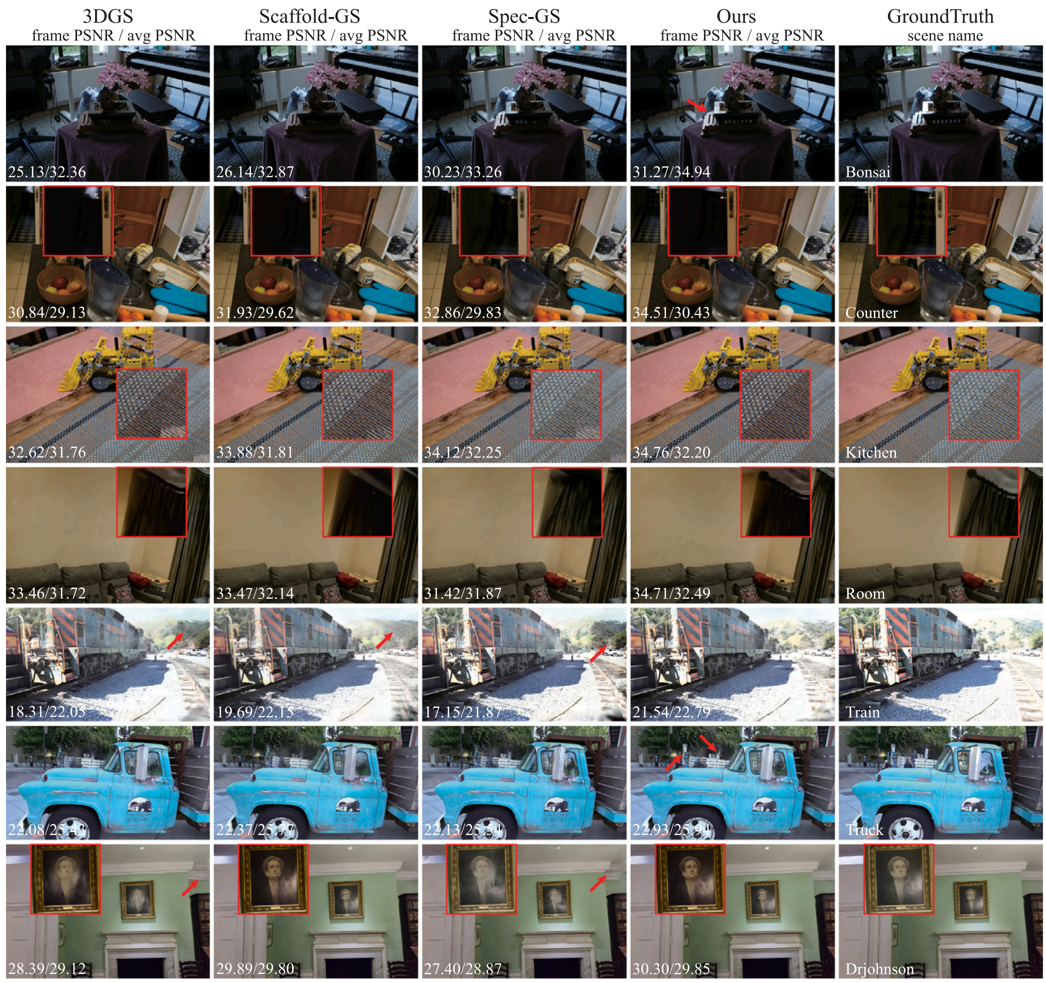


**Fig. 5.** Qualitative comparison with prior methods on several datasets and their corresponding real images. To facilitate clearer observation, we use arrows and red boxes to highlight visual differences. Our method demonstrates superior performance across multiple challenging scenes, e.g. lighting effects (Bonsai and Counter in the Mip-NeRF360 dataset [12]), fine details (Kitchen and Room in the Mip-NeRF360 dataset), under-observed situations (Train and Truck in the Tanks&Temples dataset [61]), and texture-less areas (Drjohnson in the Deepblending dataset [62]).


**Fig. 6.** Qualitative comparison with prior methods on the Synthetic Blender dataset. Significant differences in lighting effects are highlighted within the red box.

**Table 1**
Quantitative comparison with prior methods on three real-world datasets. Competitive metrics are derived from the literature, where * indicates results retrained on our device.

| Methods | Mip-NeRF360 | | | Tanks&Temples | | | Deepblending | | |
|---|---|---|---|---|---|---|---|---|---|
| | PSNR↑ | SSIM↑ | LPIPS↓ | PSNR↑ | SSIM↑ | LPIPS↓ | PSNR↑ | SSIM↑ | LPIPS↓ |
| MipNeRF360 [12] | 24.69 | 0.792 | 0.237 | 22.22 | 0.759 | 0.257 | 29.40 | 0.901 | 0.245 |
| 3DGS* [8] | 27.78 | 0.826 | 0.203 | 23.74 | 0.849 | 0.179 | 29.64 | 0.905 | 0.248 |
| Mip-Splatting [33] | 27.61 | 0.816 | 0.215 | 23.96 | 0.856 | 0.171 | 29.56 | 0.901 | 0.243 |
| VDGS [11] | 27.64 | 0.813 | 0.220 | 24.02 | 0.851 | 0.176 | 29.54 | 0.903 | 0.243 |
| Scaffold-GS* [34] | 27.98 | 0.824 | 0.207 | 23.96 | 0.853 | 0.177 | 30.21 | 0.906 | 0.254 |
| NeRF-GS [53] | 28.32 | 0.817 | 0.210 | 24.44 | 0.860 | 0.161 | 30.70 | 0.912 | 0.237 |
| Spec-GS* [10] | 27.99 | 0.834 | 0.176 | 23.70 | 0.854 | 0.166 | 29.41 | 0.904 | 0.242 |
| Ours | 28.57 | 0.834 | 0.196 | 24.37 | 0.857 | 0.166 | 30.29 | 0.911 | 0.243 |

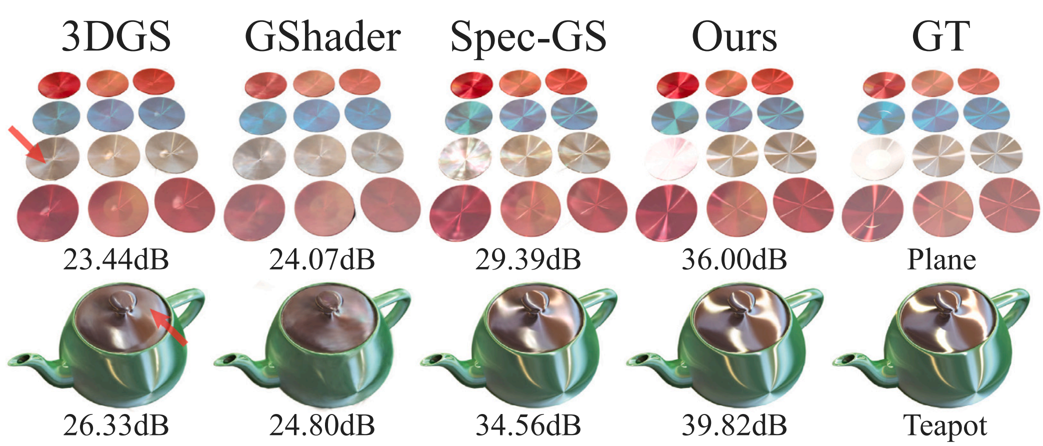


**Fig. 7.** Compared with 3DGS [8], GShader [54], and Spec-GS [10], our proposed method achieves higher PSNR scores, and visual comparison results demonstrate a significant enhancement in its ability to represent specular highlights and anisotropic effects. The arrows in the figure clearly highlight the limitations of 3DGS in capturing high-frequency scene details.

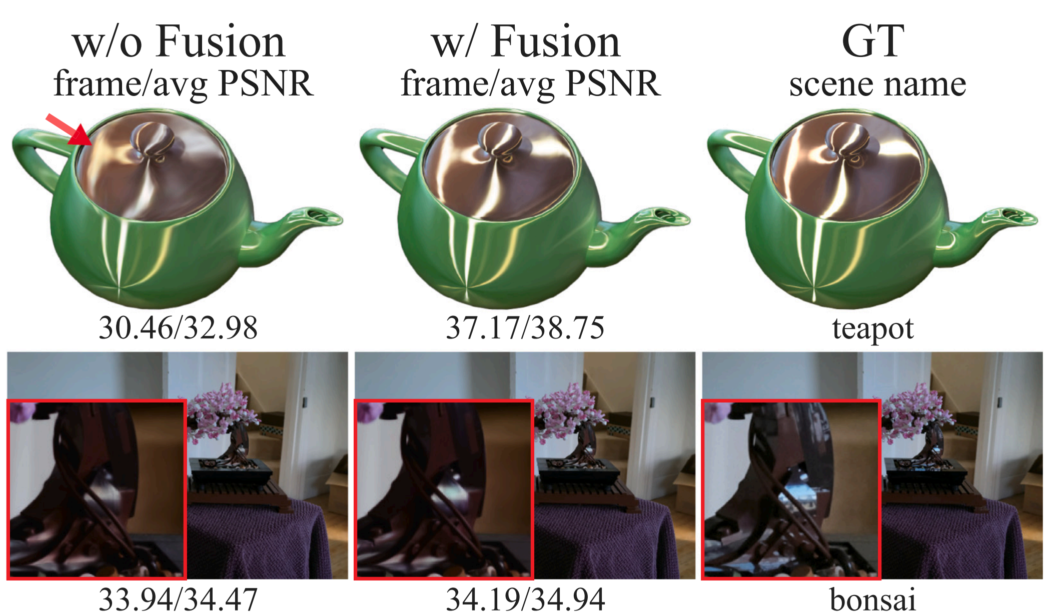


**Fig. 8.** Comparison of visual quality before and after using the spatially adaptive spherical harmonic features.

**Table 2**
Performance comparison. Rendering speed and storage size of all methods are measured on our device. Among them, the hash grid and the small MLP in our method occupy a total of 50MB of fixed storage.

| Methods | Mip-NeRF360 | | Tanks&Temples | | Deepblending | |
|---|---|---|---|---|---|---|
| | FPS | Mem | FPS | Mem | FPS | Mem |
| 3DGS* [8] | 127 | 741MB | 154 | 433MB | 126 | 667MB |
| Scaffold-GS* [34] | 91 | 203MB | 150 | 90MB | 157 | 64MB |
| Spec-GS* [10] | 40 | 847MB | 47 | 563MB | 32 | 793MB |
| Ours | 79 | 127MB | 62 | 114MB | 116 | 124MB |

#### *4.3.1. SASHF*

As shown in Table 5, SASHF significantly improves the visual quality in real-world indoor scenes and Anisotropic Synthetic datasets. Fig. 8 provides a qualitative comparison that further indicates these features' significant contribution to modeling specular highlights and anisotropic effects. Compared with Spec-GS, our method provides a more concise representation and do not require the introduction of multiple ASG functions or additional normal constraints [10].

To evaluate the effectiveness of the spatially adaptive spherical harmonic features design, we conduct a comparative analysis with alternative feature fusion strategies: the default feature concatenation used in NeRF, the Hadamard product, and the self-attention mechanism. The results in Table 6 show that these methods all fail to effectively enhance the model's capacity to capture high-frequency information. Although the Hadamard product and the self-attention mechanism have adaptive learning capabilities, they cannot construct high-dimensional feature representations. Notably, the self-attention mechanism even leads to performance degradation. Furthermore, ablation experiments indicate that increasing the spherical harmonic order (8-SH) or the number of MLP layers (4-layers) without spatially adaptive spherical harmonic features does not yield satisfactory improvements. This further highlights the irreplaceable role of this design in modeling high-frequency visual details.

#### *4.3.2. Frequency-driven adaptive activation strategy*

As shown in Table 5, omitting the frequency-driven adaptive activation strategy in real-world outdoor scenes—that is, uniformly applying SASHF to all Gaussians—leads to degraded visual quality in novel view synthesis. This degradation stems from the sparse observation of certain Gaussians, which appear in only a few training views and have a low occurrence frequency. Under such conditions, forcing SASHF on all Gaussians induces local overfitting and degrades the overall reconstruction quality. These findings motivate a selective adaptation strategy of enabling or disabling SASHF for each Gaussian based on its occurrence frequency, thereby substantially improving the method's robustness and generalization in complex outdoor environments. In contrast, Gaussians in real-world indoor scenes and synthetic datasets typically have high occurrence frequencies, so the frequency-driven adaptation yields only marginal improvements in visual quality.

Fig. 9 shows our systematic evaluation of the impact of the view-dependent criterion threshold, $\theta$, on the quality of new view synthesis. Across three representative scenes (Bicycle, Bonsai, and Playroom), the PSNR first increases and then decreases as $\theta$ increases. The optimal PSNR is achieved at $\theta = 0.3$ (marked by the red vertical dashed line) across all scenes. At this threshold, $\theta = 0.3$, Gaussians that are strongly view-dependent are effectively discriminated from those that exhibit near-isotropic behavior. Consequently, SASHF is enabled adaptively for the former and disabled for the latter, thus enhancing the visual quality of the scene.

Furthermore, eliminating the direction variable entirely results in PSNR scores of 26.93 dB for Bicycle, 31.92 dB for Bonsai, and 31.18 dB for Playroom. These results reveal a clear hierarchy of view dependency across scenes. Bonsai exhibits the strongest view dependency, evidenced by a substantial 4.68 dB PSNR decline upon direction removal, followed by Bicycle (1.01 dB). Playroom shows negligible dependency (0.01 dB), demonstrating its near-isotropic appearance. Next, we compare fixed-threshold configurations: $\theta = 0.0$ (SASHF is always enabled) and $\theta = 1.0$ (SASHF is always disabled). In Bonsai, SASHF yields a measurable gain of 0.55 dB (increasing from 35.18 to 35.73). In contrast, in Playroom, SASHF introduces modeling redundancy, causing a 0.58 dB decrease (decreasing from 31.06 to 30.48). In summary, these ablations demonstrate that the impact of SASHF is directly governed by the view dependency of the scene. Improvements are pronounced where view dependency is strong, whereas performance deteriorates where it is weak. This inverse sensitivity highlights the necessity of the frequency-driven adaptive activation strategy.

#### *4.3.3. View-dependent opacity*

As shown in Table 5, view-dependent opacity has a limited overall contribution to anisotropic illumination modeling. However, it has certain advantages in specific scenes. Fig. 10 shows that, in the Mic scene, PSNR decreases slightly (from 36.60 dB to 36.00 dB) when view-dependent opacity is disabled. Meanwhile, the results show that incorporating SASHF into the opacity calculation (O-SASHF) does not improve visual quality.

#### *4.3.4. Multi-view information smoothing*

Table 5 and Fig. 1 demonstrate that multi-view information smoothing helps prevent the model from converging to local optima, enhances generalization to unseen viewpoints, and significantly improves the overall visual quality of the scene. As shown in Fig. 11, we further evaluate the influence of different hyperparameter values $\epsilon$ on model generalization. The experimental results show that as $\epsilon$ increases, the PSNR curve on the test set gradually rises, the generalization capability steadily improves, and the scene visual quality correspondingly improves. The model achieves optimal comprehensive performance across all test images is achieved when $\epsilon = 0.99$. The case with $\epsilon = 1.0$ was not evaluated because this value disables cross-view information fusion in Eq. (7), causing the multi-view information smoothing mechanism to become ineffective. Moreover, Table 7 confirms the effectiveness of the proposed smoothing method on the vanilla 3DGS.

#### *4.3.5. Adaptive densification strategy*

Table 5 shows that the adaptive densification strategy significantly improves visual quality.Table 8 further evaluates the influence of different hyperparameter values $\beta$ on visual quality, rendering speed, and the number of Gaussians. The results show that as $\beta$ decreases, the total number of Gaussians gradually increases, visual quality improves correspondingly, and rendering speed progressively decreases. Considering the trade-off among visual quality, efficiency, and Gaussian count, we selected $\beta = 0.8$ to achieve an optimal balance. Furthermore, compared to the baseline without adaptive densification, both our method and the vanilla 3DGS benefit from this strategy by effectively reducing Gaussian density in low-frequency observation regions while increasing it in high-frequency regions, thereby enhancing detail modeling capability and overall visual quality without substantially increasing the total number of Gaussians. Table 7 further confirms the effectiveness of the adaptive densification strategy on the vanilla 3DGS.

Under the condition of fixing the hyperparameter $\beta$ to 0.8, the initial value in Eq. (8) is set to the gradient of the initial view space position to construct the ablation experiment (Ours-Init). The results show that with a non-zero initial value, the number of Gaussians in the low-frequency and high-frequency observation regions is nearly identical to the number obtained without the adaptive densification strategy. This indicates that setting the initial value to zero is crucial for enabling adaptive densification in this method.

**Table 3**
Quantitative comparison with prior methods on the Synthetic Blender and Anisotropic Synthetic datasets. To ensure fairness, we retrained Scaffold-GS and Spec-GS on the Synthetic Blender dataset with a white background. * indicates results retrained on our device.

| Methods | Synthetic blender | | | Anisotropic synthetic | | |
|---|---|---|---|---|---|---|
| | PSNR↑ | SSIM↑ | LPIPS↓ | PSNR↑ | SSIM↑ | LPIPS↓ |
| 3DGS* [8] | 33.32 | 0.969 | 0.031 | 33.82 | 0.966 | 0.062 |
| Mip-Splatting [33] | 33.54 | 0.970 | 0.029 | – | – | – |
| VDGS [11] | 33.37 | 0.969 | 0.032 | – | – | – |
| NeRF-GS [53] | 33.71 | 0.970 | 0.032 | – | – | – |
| Scaffold-GS* [34] | 33.38 | 0.968 | 0.032 | 35.34 | 0.972 | 0.052 |
| GShader* [54] | 31.48 | 0.960 | 0.042 | 30.57 | 0.948 | 0.087 |
| Spec-GS* [10] | 33.68 | 0.969 | 0.028 | 37.78 | 0.979 | 0.041 |
| Ours | 34.47 | 0.972 | 0.027 | 39.68 | 0.984 | 0.036 |

**Table 4**
Quantitative comparison with prior methods on the Shiny Blender datasets. * indicates results retrained on our device.

| Methods | Shiny blender | | | | | | |
|---|---|---|---|---|---|---|---|
| | Car | Ball | Helmet | Teapot | Toaster | Coffee | Avg. |
| PSNR↑ | | | | | | | |
| 3DGS [8] | 27.24 | 27.69 | 28.32 | 45.68 | 20.99 | 32.32 | 30.37 |
| GShader [54] | 27.90 | 30.98 | 28.32 | 45.86 | 26.21 | 32.39 | 31.94 |
| GShader* [54] | 27.51 | 29.02 | 28.73 | 43.05 | 22.86 | 31.34 | 30.42 |
| 3iGS [55] | 27.52 | 26.82 | 28.08 | 46.05 | 22.71 | 32.64 | 30.64 |
| Ours | 27.61 | 29.13 | 28.65 | 46.18 | 23.31 | 32.36 | 31.21 |
| SSIM↑ | | | | | | | |
| 3DGS [8] | 0.930 | 0.937 | 0.951 | 0.996 | 0.895 | 0.971 | 0.947 |
| GShader [54] | 0.931 | 0.965 | 0.950 | 0.996 | 0.929 | 0.971 | 0.957 |
| GShader* [54] | 0.930 | 0.954 | 0.955 | 0.995 | 0.900 | 0.969 | 0.951 |
| 3iGS [55] | 0.930 | 0.933 | 0.951 | 0.996 | 0.909 | 0.972 | 0.948 |
| Ours | 0.931 | 0.948 | 0.952 | 0.997 | 0.916 | 0.972 | 0.953 |
| LPIPS↓ | | | | | | | |
| 3DGS [8] | 0.047 | 0.161 | 0.079 | 0.007 | 0.126 | 0.078 | 0.083 |
| GShader [54] | 0.045 | 0.121 | 0.076 | 0.007 | 0.079 | 0.078 | 0.068 |
| GShader* [54] | 0.045 | 0.148 | 0.088 | 0.012 | 0.111 | 0.085 | 0.082 |
| 3iGS [55] | 0.045 | 0.166 | 0.073 | 0.006 | 0.098 | 0.077 | 0.077 |
| Ours | 0.044 | 0.137 | 0.068 | 0.006 | 0.087 | 0.077 | 0.070 |

**Table 5**
PSNR score for ablation studies.

| Methods | Mip-NeRF360 | | Anisotropic synthetic |
|---|---|---|---|
| | Outdoor | Indoor | |
| w/o V-Opacity | 25.33 | 32.40 | 39.42 |
| w/o M-Smoothing | 25.28 | 32.31 | 38.90 |
| w/o A-Densify | 25.16 | 32.26 | 39.03 |
| w/o SASHF | 25.37 | 32.34 | 37.22 |
| w/o A-Activation | 25.22 | 32.48 | **39.68** |
| Full | **25.41** | **32.51** | **39.68** |

**Table 6**
An ablation study of the spatially adaptive spherical harmonic features construction method on the Teapot scene of the Anisotropic Synthetic dataset.

| Methods | PSNR↑ | SSIM↑ | LIPIS↓ |
|---|---|---|---|
| Feature Concatenation (FC) | 32.37 | 0.982 | 0.027 |
| Hadamard Product | 32.65 | 0.984 | 0.025 |
| Self-Attention | 28.38 | 0.970 | 0.043 |
| FC (w/8-SH) | 32.81 | 0.983 | 0.026 |
| FC (w/4-layers MLPs) | 32.64 | 0.983 | 0.026 |
| Ours | **38.78** | **0.993** | **0.013** |

**Table 7**
Verification of the effectiveness of multi-view information smoothing and adaptive densification strategy in vanilla 3DGS.

| Methods | Mip-NeRF360 (PSNR↑) |
|---|---|
| 3DGS | 27.78 |
| 3dgs (w/ A-Densify) | 27.94 |
| 3dgs (w/ A-Densify and M-Smoothing) | **28.03** |

## 5. Conclusions and limitations

In this study, we propose Aniso-GS, a novel method that integrates 3DGS fast rasterization with anisotropic appearance fields. The method enhances the ability to capture high-frequency scene details by introducing spatially adaptive spherical harmonic features, effectively extending the modeling capacity of 3DGS in complex scenes characterized by specular highlights and anisotropic effects. Additionally, we introduce a multi-view information smoothing strategy to improve generalization to unseen viewpoints. We also propose an adaptive densification strategy that suppresses Gaussian density in low-frequency regions while enhancing it in high-frequency regions to improve visual quality. Experimental results demonstrate that our

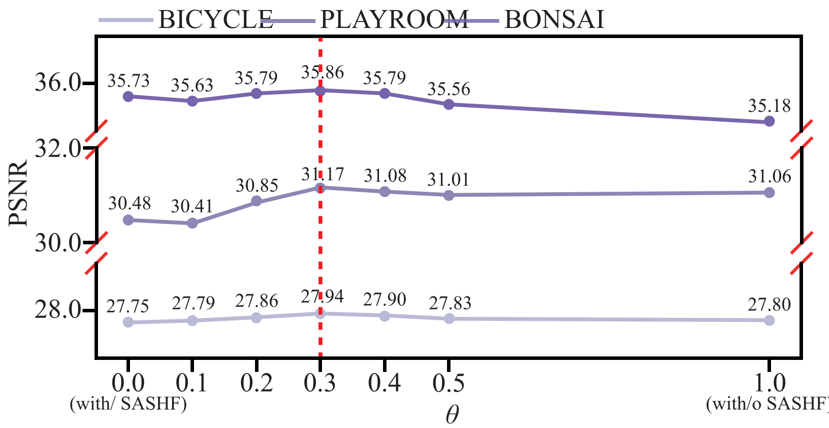


**Fig. 9.** PSNR score curves on the test set for the outdoor scene Bicycle (downsampled by a factor of 8), the indoor scene Bonsai (downsampled by a factor of 4) from the Mip-NeRF360 dataset, and the scene Playroom (downsampled by a factor of 2) from the DeepBlending dataset. By systematically varying the view-sensitivity threshold $\theta$, we analyze its impact on visual quality. Here, $\theta = 0.0$ indicates that SASHF is enabled for all Gaussians; $\theta = 1.0$ indicates that SASHF is fully disabled.

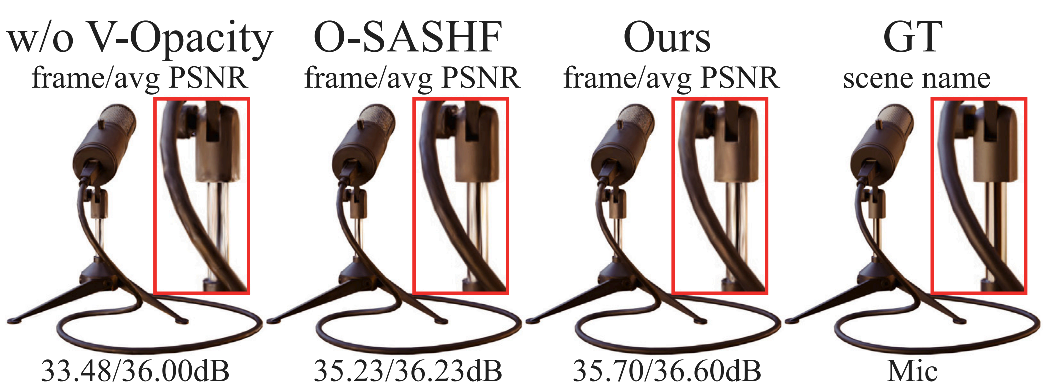


**Fig. 10.** Quantitative and qualitative comparisons of the Mic scene from the Synthetic Blender dataset. "V-Opacity" denotes view-dependent opacity, and "O-SASHF" denotes incorporating spatially adaptive spherical harmonic features into the opacity calculation. Significant visual differences are highlighted within the red box.

**Table 8**
Influence of different weight parameters $\beta$ in the adaptive densification strategy on visual quality, rendering speed, and number of 3D Gaussians. In this table, $N_{out}$ represents the number of Gaussians in the low-frequency observation regions that appear fewer than 30 times in the training set, and $N_{in}$ represents the number of Gaussians in the low-frequency observation regions that appear 30 times or more.

| Methods | $\beta$ | Bicycle | | | | | Bonsai | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | PSNR↑ | FPS | $N_{out}$ | $N_{in}$ | $N_{total}$ | PSNR↑ | FPS | $N_{out}$ | $N_{in}$ | $N_{total}$ |
| 3dgs | – | 25.69 | 72 | 3.76M | 1.99M | 5.75M | 32.36 | 192 | 0.38 | 0.87M | 1.25M |
| 3dgs (w/ A-Densify) | 0.8 | 25.76 | 71 | 3.04M | 2.47M | 5.51M | 32.61 | 167 | 0.24M | 1.06M | 1.30M |
| Ours (w/o A-Densify) | – | 25.82 | 66 | 2.31M | 1.26M | 3.57M | 34.54 | 120 | 0.27M | 0.62M | 0.89M |
| Ours-Init | 0.8 | 25.78 | 72 | 2.33M | 1.27M | 3.60M | 34.50 | 125 | 0.25M | 0.65M | 0.90M |
| Ours | 0.9 | 25.85 | 64 | 1.10M | 1.39M | 2.49M | 34.77 | 126 | 0.11M | 0.63M | 0.74M |
| | 0.8 | 25.97 | 55 | 1.94M | 1.68M | 3.62M | 34.94 | 103 | 0.18M | 0.79M | 0.97M |
| | 0.7 | 26.00 | 52 | 2.54M | 1.90M | 4.44M | 34.90 | 90 | 0.23M | 0.91M | 1.14M |
| | 0.5 | 26.06 | 44 | 3.48M | 2.34M | 5.82M | 34.96 | 79 | 0.31M | 1.14M | 1.45M |
| | 0.2 | 26.08 | 38 | 4.81M | 3.00M | 7.81M | 35.02 | 64 | 0.42M | 1.51M | 1.93M |

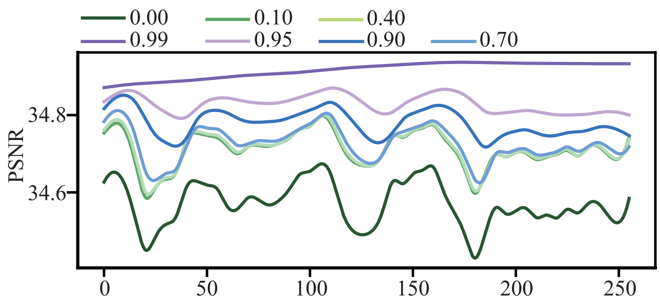


**Fig. 11.** Taking the Bonsai in the Mip-NeRF360 dataset as an example, the influence of adjusting different weight parameter $\epsilon$ on the generalization of the model is analyzed.

method demonstrates superior performance in rendering specular highlights and anisotropic effects, and achieves advanced visual quality in general scenes.

However, rendering speed is limited by the number of visible Gaussians per view, since their opacity and color are generated by the MLPs. Future research should investigate methods to compress redundant Gaussians to enhance rendering efficiency.

## CRediT authorship contribution statement

**Gang Li:** Writing – original draft, Software, Methodology, Conceptualization. **Zhongliang Fu:** Writing – review & editing, Project administration, Conceptualization. **Zhao Liu:** Writing – review & editing, Visualization, Validation, Data curation. **Shengyuan Zhang:** Supervision, Resources, Investigation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The numerical calculations in this paper have been done on the supercomputing system in the Supercomputing Center of Wuhan University.

## Data availability

The code is available at https://github.com/liganggis/Aniso-GS. And the data used in this work is publicly available.

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